Home / Grade 10 / Algebra / Quarter 2

Rational equations

Clear the denominators, solve, then throw away the roots the original equation never allowed.

LESSON 28–29 | 2 × 40 MIN | ALGEBRA 10, §2.1 | P1 · 1.7 (EXTENSION)

LESSON CLOCK

8' | 16' | 16' | 18' | 22' | 10'

Why a check is compulsory here	8 min
The method	16 min
Extraneous roots	16 min
Disguised quadratics	18 min
Practice	22 min
Homework	10 min

LEARNING OBJECTIVES

- State the domain of a rational equation before solving it.
- Solve an equation by finding the lowest common denominator.
- Identify and reject extraneous roots.
- Solve equations that become quadratic after a substitution.

TEXTBOOK

National	pp. 94–103
Cambridge	Pure Mathematics 1, pp. 18–23
Workbook	P1 Exercise 1F

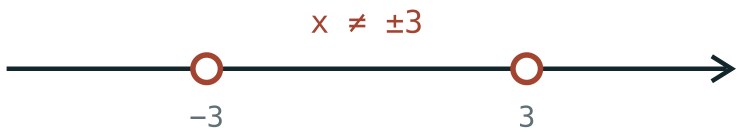
01 Explanation

The domain comes first

WRITE IT BEFORE YOU SOLVE

Every denominator gives a forbidden value. Write $x \neq \dots$ on the first line, before any algebra. It is not a formality: it is the only thing that will tell you which of your final answers is real.

For $\dfrac{3}{x-2} + \dfrac{1}{x} = 2$ the domain is $x \neq 0$ and $x \neq 2$.



The two excluded points, marked as open circles before any solving begins.

The method

1. State the domain.
2. Factorise every denominator, and find the lowest common denominator.
3. Multiply every term by the LCD, and cancel.
4. Solve the polynomial equation that results.
5. Reject any root that is outside the domain.

$$\frac{3}{x-2} + \frac{1}{x} = 2 \Rightarrow 3x + (x - 2) = 2x(x - 2)$$

$$4x - 2 = 2x^2 - 4x \Rightarrow 2x^2 - 8x + 2 = 0 \Rightarrow x^2 - 4x + 1 = 0 \Rightarrow x = 2 \pm \sqrt{3}$$

Both roots are allowed, so both are answers.

MULTIPLY **EVERY** TERM

A term with no fraction is multiplied by the LCD too. Forgetting the right-hand side is the commonest arithmetic error in the topic.

Extraneous roots — where they come from

Multiplying by $x - 3$ is only reversible when $x \neq 3$. If it happens to be zero, you have multiplied both sides by zero, and the new equation is **not equivalent** to the old one — it has extra solutions the original never had.

$$\frac{x^2}{x-3} = \frac{9}{x-3} \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$$

But $x = 3$ is outside the domain. The only solution is $x = -3$.

THE CHECK IS PART OF THE ANSWER

Never write a final answer without testing it against the domain line you wrote at the start. Examiners award a mark for the rejection, and take one for its absence.

Disguised quadratics

When the same expression appears twice, name it and the equation becomes quadratic:

$$x^2 + \frac{1}{x^2} = 2.5 \text{ , let } t = x + \frac{1}{x}$$

Then $t^2 = x^2 + 2 + \frac{1}{x^2}$, so the equation reads $t^2 - 2 = 2.5$, giving $t = \pm 1.5$ — two much easier equations to finish.

The other standard shape is $\frac{x}{x+1} + \frac{x+1}{x} = 2.5$: set $t = \frac{x}{x+1}$ and it becomes $t + \frac{1}{t} = 2.5$.

02 Worked examples

EXAMPLE 1 **Solve $\frac{2}{x} + \frac{3}{x+1} = 2$.**

1. Domain $x \neq 0, -1$.
2. LCD $x(x + 1)$.

   $2(x + 1) + 3x = 2x(x + 1)$
3. $5x + 2 = 2x^2 + 2x$
4. $2x^2 - 3x - 2 = 0 \Rightarrow (2x + 1)(x - 2) = 0$

   Both allowed.

ANSWER

$x = 2$ or $x = -0.5$

EXAMPLE 2 **Solve $\frac{x}{x-1} - \frac{1}{x-1} = 3$.**

1. Domain $x \neq 1$.

2. Same denominator: $\frac{x-1}{x-1} = 1$.

   For every allowed x.

3. $1 = 3$

   A contradiction.

ANSWER

No solution

EXAMPLE 3 **Solve $\frac{x^2}{x-2} = \frac{4}{x-2}$.**

1. Domain $x \neq 2$.

2. $x^2 = 4 \Rightarrow x = \pm 2$

3. $x = 2$ is excluded.

ANSWER

$x = -2$ only

03 Interactive model

Show the graph of each side and let the class see where the removed point sits.

Domain first

Domain of $\frac{1}{x-5} = 2$:

- $x \neq 0$
- $x \neq 5$
- $x > 5$
- $\mathbb{R}$

Solving $\frac{x^2}{x-3} = \frac{9}{x-3}$ gives:

- $x = \pm 3$
- $x = 3$
- $x = -3$
- no solution

An extraneous root appears because:

- of arithmetic error
- we multiplied by something that can be zero
- the equation is wrong
- it never does

Solve $\frac{2}{x} = 1$:

- $x = 2$
- $x = 0.5$
- $x = 0$
- no solution

For $\frac{1}{x} + \frac{1}{x-1} = 5$ the LCD is:

- x
- $x - 1$
- $x(x - 1)$
- x^2

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rational equation	Ratsional tenglama	Рациональное уравнение
Denominator	Maxraj	Знаменатель
Lowest common denominator	Eng kichik umumiy maxraj	Наименьший общий знаменатель
Extraneous root	Chet ildiz	Посторонний корень
Domain restriction	Aniqlanish sohasi cheklovi	Ограничение области определения
Clearing fractions	Kasrlardan qutulish	Освобождение от дробей
Equivalent equations	Teng kuchli tenglamalar	Равносильные уравнения
Check (substitution)	Tekshirish	Проверка
Substitution variable	Yordamchi o‘zgaruvchi	Вспомогательная переменная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first line of a rational equation should be:

- the LCD
- the domain
- the answer
- a graph

Multiplying by the LCD may introduce:

- lost roots
- extraneous roots
- nothing
- fractions

$\frac{x-1}{x-1}$ equals 1:

- always
- for $x \neq 1$
- never
- only at $x = 1$

For $x^2 + \frac{1}{x^2} = 2.5$ a good substitution is:

- $t = x^2$
- $t = x + \frac{1}{x}$
- $t = x - 1$
- none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Domain of $\frac{1}{x-4} = 3$

ANSWER $x \neq 4$

2. Solve $\frac{6}{x} = 2$

ANSWER $x = 3$

3. Solve $\frac{1}{x+1} = 1$

ANSWER $x = 0$

4. Solve $\frac{x}{2} = \frac{3}{x}$

ANSWER $x = \pm\sqrt{6}$

5. Domain of $\frac{1}{x} + \frac{1}{x-2}$

ANSWER $x \neq 0, 2$

6. LCD of $\frac{1}{x}$ *and* $\frac{1}{x+3}$

ANSWER $x(x+3)$

7. Solve $\frac{4}{x-1} = 2$

ANSWER $x = 3$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{2}{x} + \frac{3}{x+1} = 2$

ANSWER $x = 2, -0.5$

2. Solve $\frac{x}{x+2} = \frac{3}{x}$

ANSWER $x = 6$ or $x = -1$

3. Solve $\frac{1}{x-1} + \frac{1}{x+1} = 1$

ANSWER $x = 1 \pm \sqrt{3}$

4. Solve $\frac{x^2}{x-2} = \frac{4}{x-2}$

ANSWER $x = -2$

5. Solve $\frac{3}{x} - \frac{2}{x^2} = 1$

ANSWER $x = 1, 2$

6. Solve $\frac{x+1}{x-1} = \frac{x-1}{x+1}$

ANSWER $x = 0$

7. Solve $\frac{x}{x-3} - \frac{3}{x-3} = 2$

ANSWER no solution

Hard · stretch and reason

7 PROBLEMS

1. Solve $x^2 + \dfrac{1}{x^2} = 2.5$

ANSWER $x = \pm 2, \pm 0.5$

2. Solve $\dfrac{x}{x+1} + \dfrac{x+1}{x} = 2.5$

ANSWER $x = 1, -2$

3. Solve $\dfrac{1}{x^2-1} + \dfrac{1}{x-1} = 1$

ANSWER $x = 2$ or $x = -1$ rejected

4. Solve $\dfrac{2x}{x^2-4} + \dfrac{1}{x-2} = \dfrac{3}{x+2}$

ANSWER $x = -8$

5. Find k so $\dfrac{x}{x-2} = k$ has no solution

ANSWER $k = 1$

6. Solve $\dfrac{x^2-4}{x^2-2x} = 1$

ANSWER no solution

7. Solve $(x + \dfrac{1}{x})^2 - 4(x + \dfrac{1}{x}) + 3 = 0$

ANSWER $x = 1$ (from $t = 3$: $\dfrac{3 \pm \sqrt{5}}{2}$)

07 Homework

Homework — 5 tasks

Every solution must open with the domain and close with the check.

1. Solve $\dfrac{3}{x} + \dfrac{2}{x-1} = 3$.

2. Solve $\dfrac{x}{x-4} = \dfrac{2}{x+1}$.

3. Solve $\dfrac{x^2}{x+3} = \dfrac{9}{x+3}$ and say which root is rejected and why.

4. Solve $\dfrac{1}{x} + \dfrac{1}{x+2} = \dfrac{3}{4}$.

5. Solve $x^2 + \dfrac{4}{x^2} = 5$ using a substitution.

Systems of rational equations

Two rational equations at once — substitution, symmetric sums, and the domain that must survive both.

LESSON 30–31 | 2 × 40 MIN | ALGEBRA 10, §2.2 | P1 · 1.7

LESSON CLOCK

8' | 20' | 22' | 16' | 20' | 4

What is new	8 min
Substitution	20 min
Symmetric systems	22 min
Graphical meaning	16 min
Practice	20 min
Homework	4 min

LEARNING OBJECTIVES

- Solve a system containing rational equations by substitution.
- Use the sum-and-product substitution for symmetric systems.
- State the combined domain of a system.
- Interpret the solutions as points of intersection.

TEXTBOOK

National	pp. 104–113
Cambridge	Pure Mathematics 1, pp. 18–23
Workbook	P1 Exercise 1F

01 Explanation

The combined domain

BOTH EQUATIONS MUST ACCEPT THE ANSWER

A pair (x, y) solves the system only if it lies in the domain of **both** equations. So the excluded values of each are excluded from the whole system.

Everything else is the ordinary technique of Grade 9 — eliminate a variable, solve, substitute back — with the extra step of clearing denominators first.

Substitution

When one equation gives a variable cheaply, use it:

$$x + y = 5 \text{ and } \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$$

The second equation is $\frac{x+y}{xy} = \frac{5}{6}$, so $\frac{5}{xy} = \frac{5}{6}$ and $xy = 6$. With $x + y = 5$ the numbers are the roots of $t^2 - 5t + 6 = 0$.

$$(x, y) = (2, 3) \text{ or } (3, 2)$$

> TWO ORDERED PAIRS, NOT ONE
>
> A symmetric system almost always gives both orders. Writing only one loses half the answer.

Symmetric systems

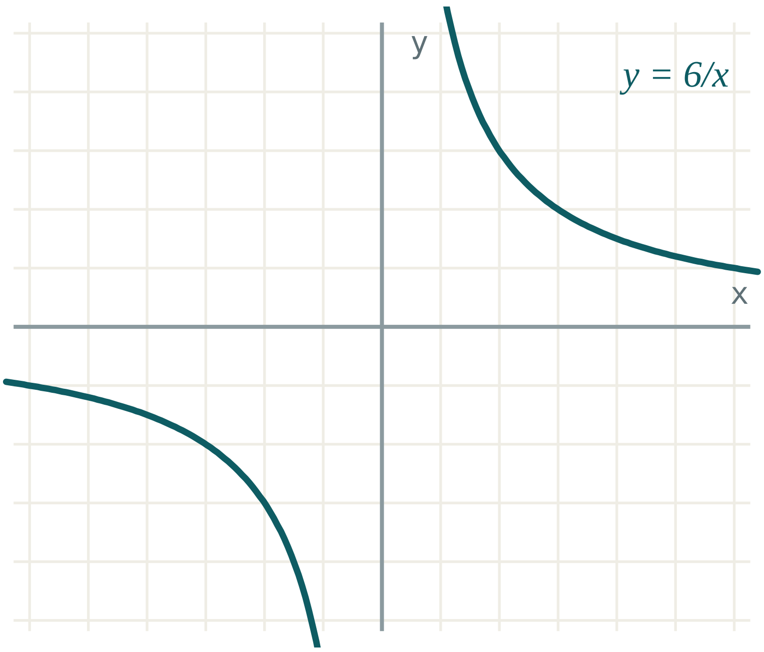
A system is **symmetric** when swapping x and y leaves it unchanged. Every such system can be rewritten in $s = x + y$ and $p = xy$:

EXPRESSION	IN S AND P
$x^2 + y^2$	$s^2 - 2p$
$\frac{1}{x} + \frac{1}{y}$	$\frac{s}{p}$
$x^3 + y^3$	$s^3 - 3ps$
$(x - y)^2$	$s^2 - 4p$

Solve for s and p, then recover x and y as the roots of $t^2 - st + p = 0$.

What the solutions mean

Each equation is a curve. The solutions of the system are the points where the curves cross — so the number of solutions is the number of intersections.



xy = 6 is a hyperbola; x + y = 5 is a line. Two crossings, two solutions.

A system with no solution is a line missing its curve entirely; a system with one solution is a tangency.

02 Worked examples

EXAMPLE 1 **Solve $x + y = 7$, $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{7}{12}$.**

1. $\dfrac{x+y}{xy} = \dfrac{7}{12}$
2. $\dfrac{7}{xy} = \dfrac{7}{12} \Rightarrow xy = 12$
3. $t^2 - 7t + 12 = 0$
4. $t = 3, 4$

ANSWER

$(3, 4)$ and $(4, 3)$

EXAMPLE 2 **Solve $xy = 12, x^2 + y^2 = 25$.**

1. $s^2 - 2p = 25$ with $p = 12$.
   $s^2 = 49$
2. $s = \pm 7$
3. $s = 7$: $t^2 - 7t + 12 = 0 \Rightarrow 3, 4$.
4. $s = -7$: $t^2 + 7t + 12 = 0 \Rightarrow -3, -4$.

ANSWER

$(3,4), (4,3), (-3,-4), (-4,-3)$

EXAMPLE 3 **Solve $\frac{2}{x} - \frac{1}{y} = 1, \frac{1}{x} + \frac{2}{y} = 3$.**

1. Let $u = \frac{1}{x}, v = \frac{1}{y}$.
   $2u - v = 1, u + 2v = 3$
2. Solve: $u = 1, v = 1$.
3. $x = 1, y = 1$

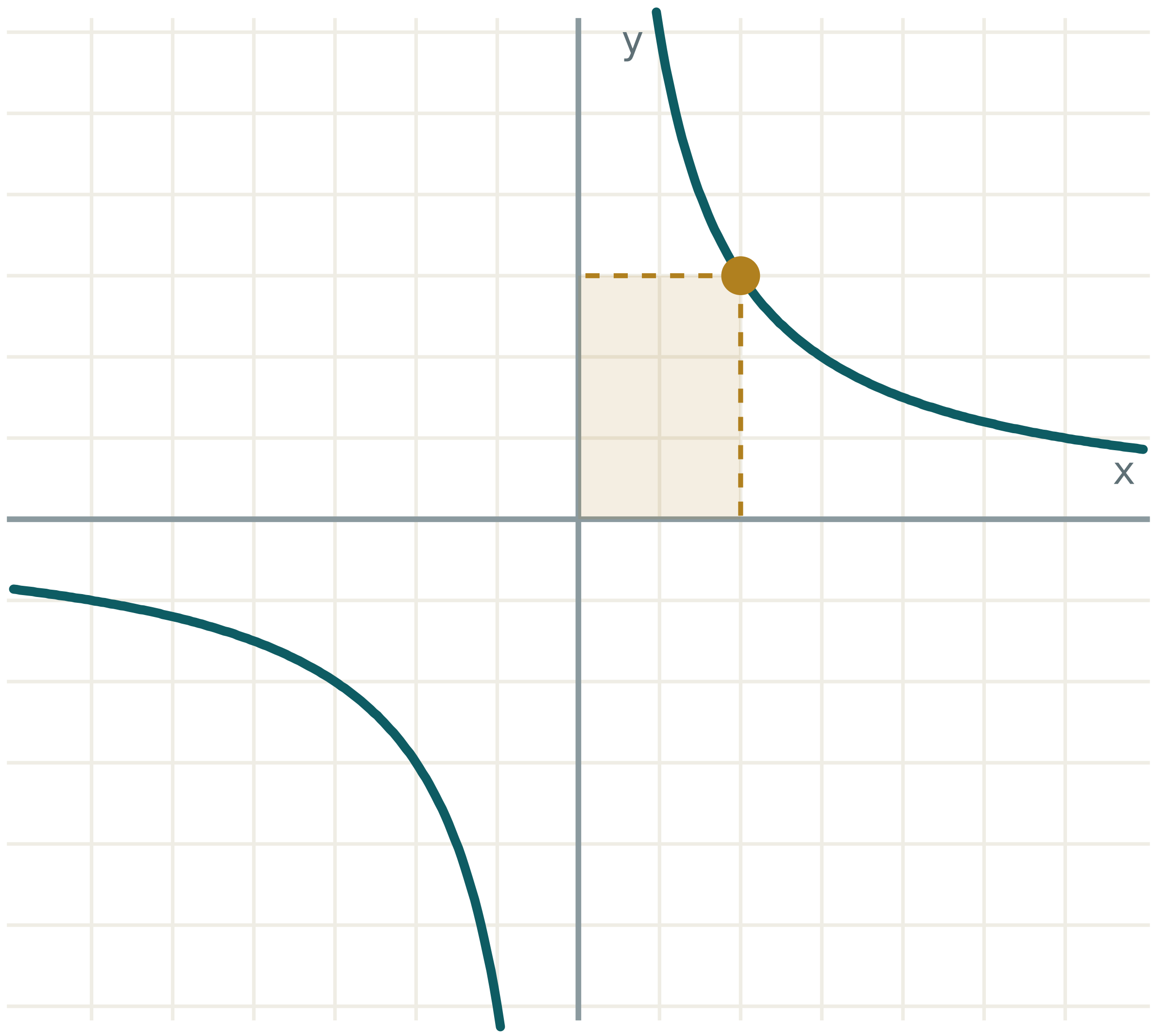
ANSWER

$(1, 1)$

03 Interactive model

Plot both curves and count the crossings before solving.

The graph of y = k ⁄ x



k **6** | x **2** | y = k/x **3** | x · y **6** | quadrants **I and III**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of equations	Tenglamalar sistemasi	Система уравнений
Substitution method	O‘rniga qo‘yish usuli	Метод подстановки
Symmetric system	Simmetrik sistema	Симметричная система
Sum and product	Yig‘indi va ko‘paytma	Сумма и произведение
Combined domain	Umumiy aniqlanish sohasi	Общая область определения
Point of intersection	Kesishish nuqtasi	Точка пересечения
Consistent system	Birgalikdagi sistema	Совместная система
Inconsistent system	Birgalikda bo‘lmagan sistema	Несовместная система

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A symmetric system usually gives:

- one pair
- two ordered pairs
- no pairs
- infinitely many

$x^2 + y^2$ in terms of s, p is:

- $s^2 + 2p$
- $s^2 - 2p$
- $s^2 - p$
- $2s - p$

For $\frac{1}{x} + \frac{1}{y}$ a good substitution is:

- $u = x, v = y$
- $u = \frac{1}{x}, v = \frac{1}{y}$
- $u = xy$
- none

The solutions of a system are:

- the roots of one equation
- the points where the curves meet
- the domains
- the asymptotes

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x + y = 5, xy = 6$

   ANSWER $(2,3), (3,2)$

2. Solve $x + y = 8, xy = 15$

   ANSWER $(3,5), (5,3)$

3. Solve $x - y = 1, xy = 6$

   ANSWER $(3,2), (-2,-3)$

4. Write $x^2 + y^2$ with s, p

   ANSWER $s^2 - 2p$

5. Write $\frac{1}{x} + \frac{1}{y}$ with s, p

   ANSWER $\frac{s}{p}$

6. Domain of $\frac{1}{x} + \frac{1}{y} = 1$

   ANSWER $x \neq 0, y \neq 0$

7. Solve $xy = 4, x = y$

   ANSWER $(2,2), (-2,-2)$

Medium · the standard question

7 PROBLEMS

1. Solve $x + y = 7,\ \frac{1}{x} + \frac{1}{y} = \frac{7}{12}$

ANSWER $(3,4), (4,3)$

2. Solve $xy = 12,\ x^2 + y^2 = 25$

ANSWER $(\pm 3,\pm 4)$ matching signs

3. Solve $\frac{2}{x} - \frac{1}{y} = 1,\ \frac{1}{x} + \frac{2}{y} = 3$

ANSWER $(1, 1)$

4. Solve $x + y = 6,\ x^2 + y^2 = 20$

ANSWER $(2,4), (4,2)$

5. Solve $x - y = 2,\ \frac{1}{x} - \frac{1}{y} = -\frac{1}{4}$

ANSWER $(4,2)$ or $(-2,-4)$

6. Solve $xy = 6,\ x + y = -5$

ANSWER $(-2,-3), (-3,-2)$

7. How many solutions has $x + y = 1,\ xy = 1$?

ANSWER none — $D < 0$

Hard · stretch and reason

7 PROBLEMS

1. Solve $x + y = 4, x^3 + y^3 = 28$

   ANSWER $(1,3), (3,1)$

2. Solve $x^2 + y^2 = 13, \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$

   ANSWER $(2,3), (3,2)$

3. Solve $\frac{x}{y} + \frac{y}{x} = \frac{5}{2}, x + y = 6$

   ANSWER $(2,4), (4,2)$

4. Solve $x + y + xy = 11, x^2 y + x y^2 = 30$

   ANSWER $(2,3), (3,2)$ or $s = 6, p = 5$

5. Find k so $x + y = 4, xy = k$ has exactly one solution

   ANSWER $k = 4$

6. Solve $\frac{1}{x} + \frac{1}{y} = \frac{1}{2}, \frac{1}{x^2} + \frac{1}{y^2} = \frac{5}{36}$

   ANSWER $(6,3), (3,6)$

7. Interpret $x + y = 5, xy = 6$ graphically

   ANSWER A line meeting a hyperbola in two points

07 Homework

Homework — 5 tasks

Write both ordered pairs whenever the system is symmetric.

1. Solve $x + y = 9, xy = 20$.
2. Solve $x + y = 5, \frac{1}{x} + \frac{1}{y} = \frac{5}{6}$.
3. Solve $x^2 + y^2 = 41, xy = 20$.

4. Solve $\dfrac{3}{x} + \dfrac{2}{y} = 4$, $\dfrac{1}{x} - \dfrac{1}{y} = 1$.

5. Explain, with a sketch, why $x + y = 1$ and $xy = 1$ have no common solution.

Rational inequalities

Never cross-multiply. Move everything to one side, combine into one fraction, and let the sign chart do the work.

LESSON 32–34 | 3 × 40 MIN | ALGEBRA 10, §2.3 | P1 · 1.2 (EXTENSION)

LESSON CLOCK

14' | 22' | 26' | 22' | 40' | 11

The forbidden step	14 min
One fraction, compared with zero	22 min
The sign chart	26 min
Repeated factors	22 min
Practice	40 min
Homework	11 min

LEARNING OBJECTIVES

- Explain why multiplying an inequality by a variable expression is forbidden.
- Reduce a rational inequality to a single fraction compared with zero.
- Build and read a sign chart, including the excluded points.
- Write the answer in interval notation.

TEXTBOOK

National	pp. 114–128
Cambridge	Pure Mathematics 1, pp. 6–11
Workbook	P1 Exercise 1C

01 Explanation

The one thing you may not do

NEVER MULTIPLY AN INEQUALITY BY $X - 3$

You do not know its sign. If it is negative the inequality reverses; if it is zero it collapses. $\frac{1}{x} > 1$ does **not** give $1 > x$ — try $x = -1$: $-1 > 1$ is false, but $1 > -1$ is true.

Multiplying by a **positive number** is safe. Multiplying by an expression whose sign you do not know is not, and there is no way to repair it afterwards.

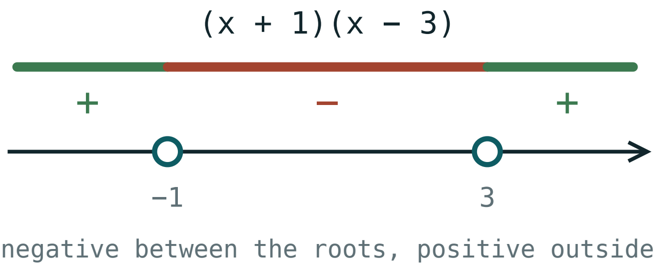
The method

FOUR STEPS, ALWAYS

1 Move everything to one side, so the other side is 0. **2** Combine into a single fraction and factorise top and bottom. **3** Mark every zero of the numerator and of the denominator on a number line. **4** Test the sign in each interval and read the answer off.

$$\frac{1}{x} > 1 \Rightarrow \frac{1}{x} - 1 > 0 \Rightarrow \frac{1-x}{x} > 0$$

Critical values 0 and 1. Testing $-1, 0.5, 2$ gives $-, +, -$, so the answer is $0 < x < 1$.



The critical values cut the line into intervals. One test point settles each.

Open or closed?

CRITICAL VALUE FROM	INEQUALITY	ENDPOINT
numerator	$\leq$ *or* $\geq$	included — closed circle
numerator	$<$ *or* $>$	excluded — open circle
denominator	any	**always** excluded

A DENOMINATOR ZERO IS NEVER INCLUDED

Even with ≥. The expression has no value there at all, so it cannot satisfy anything.

Repeated factors

The sign alternates as you cross a simple zero. It does **not** alternate at a repeated factor of even order — the graph touches the axis and turns back.

$$\frac{(x-1)^2(x+2)}{x-3} \geq 0$$

X	< -2	-2	$(-2,1)$	1	$(1,3)$	3	> 3
sign	+	0	−	0	−	×	+

Answer: $x \leq -2$, or $x = 1$, or $x > 3$. The isolated point 1 is part of the solution because the numerator is zero there and ≥ allows it.

A QUICK RULE

Start at the far right, where every factor is positive, and count the sign changes leftwards. A factor of odd power changes the sign; a factor of even power does not.

02 Worked examples

EXAMPLE 1 **Solve $\frac{x-2}{x+3} < 0$.**

1. Already one fraction against zero.
2. Critical values 2 and -3.
3. Test $-4, 0, 3$: $+, -, +$.
4. Both ends open.

ANSWER

$-3 < x < 2$

EXAMPLE 2 **Solve $\frac{2}{x-1} \le 3$.**

1. $\frac{2}{x-1} - 3 \le 0$

2. $\frac{2-3(x-1)}{x-1} = \frac{5-3x}{x-1} \le 0$

3. Critical $x = \frac{5}{3}$ (closed), $x = 1$ (open).

4. Test $0, 1.2, 2$: $-, +, -$.

ANSWER

$x < 1$ or $x \ge \frac{5}{3}$

EXAMPLE 3 **Solve $\frac{(x+1)(x-4)}{x^2} > 0$.**

1. Critical $-1, 0, 4$.
   $x = 0$ is a double root of the denominator.

2. Signs: $+, -, -, +$ across the four intervals.
   x^2 never changes sign.

3. Ends all open.

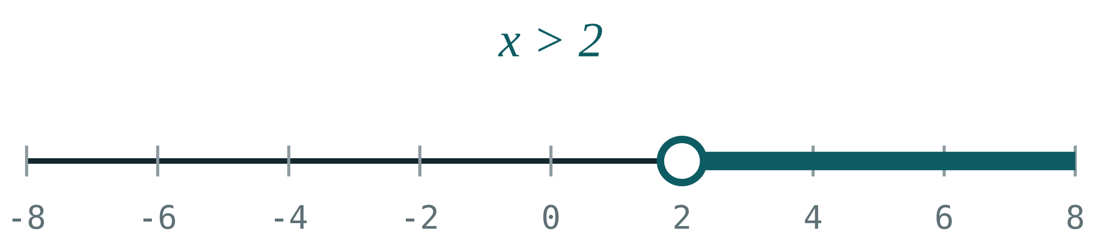
ANSWER

$x < -1$ or $x > 4$

03 Interactive model

Draw the number line first, then fill in the signs interval by interval.

Reading a sign chart



inequality `x > 2` | boundary `2 (open)` | x = 0 works? **no** | x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rational inequality	Ratsional tengsizlik	Рациональное неравенство
Sign chart (interval method)	Oraliqlar usuli	Метод интервалов
Critical value	Kritik qiymat	Критическое значение
Excluded point	Chetlatilgan nuqta	Выколотая точка
Closed circle	To‘ldirilgan nuqta	Закрашенная точка
Open circle	Bo‘sh nuqta	Выколотая точка
Interval notation	Oraliq belgilanishi	Интервальная запись
Union of intervals	Oraliqlar birlashmasi	Объединение промежутков
Sign of a factor	Ko‘paytuvchi ishorasi	Знак множителя
Multiplicity	Karralilik	Кратность

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{1}{x} > 1$ gives:

- $x < 1$
- $0 < x < 1$
- $x > 1$
- $x > 0$

A zero of the denominator is:

- always included
- always excluded
- included if ≥
- sometimes

Crossing a factor of even power the sign:

- changes
- does not change
- becomes zero
- is undefined

$\underline{x - 2}\ \ x + 3 \ < 0$ gives:

- $x < -3$
- $-3 < x < 2$
- $x > 2$
- $x < 2$

The first step of a rational inequality is:

- cross-multiply
- get zero on one side
- square it
- take reciprocals

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up `7 PROBLEMS`

1. Solve $\frac{1}{x} > 0$

ANSWER $x > 0$

2. Solve $\frac{1}{x} < 0$

ANSWER $x < 0$

3. Solve $\frac{x}{x-1} > 0$

ANSWER $x < 0$ or $x > 1$

4. Solve $\frac{x-3}{x+1} < 0$

ANSWER $-1 < x < 3$

5. Solve $(x - 1)(x + 2) > 0$

ANSWER $x < -2$ or $x > 1$

6. Is $x = 2$ allowed in $\frac{1}{x-2} \geq 0$?

ANSWER no

7. Solve $\frac{2}{x} \geq 0$

ANSWER $x > 0$

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{1}{x} > 1$

ANSWER $0 < x < 1$

2. Solve $\frac{2}{x-1} \leq 3$

ANSWER $x < 1$ or $x \geq \frac{5}{3}$

3. Solve $\frac{x+2}{x-3} \geq 0$

ANSWER $x \leq -2$ or $x > 3$

4. Solve $\frac{x}{x^2-4} > 0$

ANSWER $-2 < x < 0$ or $x > 2$

5. Solve $\frac{(x+1)(x-4)}{x^2} > 0$

ANSWER $x < -1$ or $x > 4$

6. Solve $\frac{3}{x+2} < 1$

ANSWER $x < -2$ or $x > 1$

7. Solve $\frac{x-1}{x+1} \leq 2$

ANSWER $x < -1$ or $x \geq -3$ → $x \leq -3$ or $x > -1$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\dfrac{(x-1)^2(x+2)}{x-3} \geq 0$

ANSWER $x \leq -2$, $x = 1$, or $x > 3$

2. Solve $\dfrac{x^2-5x+6}{x^2-4} \leq 0$

ANSWER $-2 < x \leq 2$ excluding 2 → $-2 < x < 2$ or $x = 3$

3. Solve $\dfrac{1}{x-1} > \dfrac{1}{x+1}$

ANSWER $x > 1$ or $-1 < x$ → $-1 < x < 1$ fails; $x > 1$ or $x < -1$

4. Solve $\dfrac{x}{x-2} \geq \dfrac{3}{x-2}$

ANSWER $x > 2$

5. Solve $\dfrac{x^3-x}{x+3} < 0$

ANSWER $x < -3$ or $-1 < x < 0$ or $x > 1$ — check signs

6. Find all k with $\dfrac{x^2+k}{x} > 0$ for every $x > 0$

ANSWER $k \geq 0$

7. Explain why $\dfrac{(x-2)^2}{x-2}$ and $x - 2$ give different solution sets

ANSWER The first excludes $x = 2$

07 Homework

Homework — 6 tasks

Every answer needs the number line drawn, with open and closed circles marked.

1. Solve $\dfrac{x-5}{x+2} < 0$.

2. Solve $\dfrac{3}{x-2} \geq 1$.

3. Solve $\dfrac{x}{x^2-9} \leq 0$.

4. Solve $\dfrac{(x-2)^2(x+1)}{x-5} \geq 0$.

5. Explain in three sentences why $\dfrac{1}{x} > 1$ is not $1 > x$.

6. Solve $\dfrac{2}{x+1} < \dfrac{1}{x}$.

Systems of rational inequalities

Solve each one alone, draw them on the same line, and keep only the overlap.

LESSON 35–37 | 3 × 40 MIN | ALGEBRA 10, §2.4 | P1 · 1.2 (EXTENSION)

LESSON CLOCK

10 | 22' | 26' | 24' | 38' | 15'

And versus or	10 min
The method	22 min
Worked systems	26 min
Empty and unbounded answers	24 min
Practice	38 min
Homework	15 min

LEARNING OBJECTIVES

- Solve each inequality of a system separately.
- Represent the solution sets on one number line and read the intersection.
- Distinguish a system (“and”) from a collection (“or”).
- Handle systems whose solution set is empty.

TEXTBOOK

National	pp. 129–142
Cambridge	Pure Mathematics 1, pp. 6–11
Workbook	P1 Exercise 1C

01 Explanation

“And” or “or”

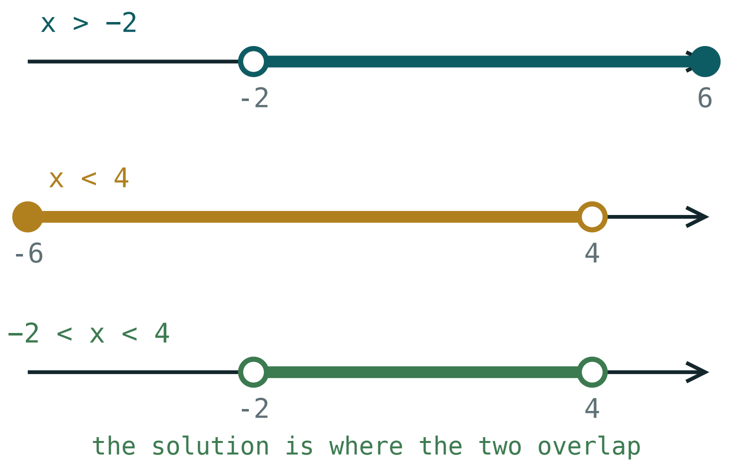
TWO DIFFERENT QUESTIONS

A **system** — written with a brace — asks for the values satisfying **every** inequality: the **intersection**. A **collection** asks for those satisfying **at least one**: the **union**.

$x > 1$ and $x < 4$ gives $1 < x < 4$. The same two as a collection give all of $\mathbb{R}$. The brace matters.

The method

1. Solve each inequality on its own, by the sign-chart method of the last lesson.
2. Draw all the solution sets on **one** number line, one above another.
3. Shade the part covered by every line.
4. Read the answer as an interval or a union of intervals.



Three solution sets drawn above one another. The answer is the strip covered by all three.

> DO NOT TRY TO SOLVE THEM SIMULTANEOUSLY
>
> Inequalities are not equations: adding or subtracting them loses information. Solve each separately, then intersect on the picture.

Worked systems

Example. $\frac{1}{x} > 0$ and $\frac{x-2}{x+1} \le 0$.

INEQUALITY	SOLUTION
$\frac{1}{x} > 0$	$x > 0$
$\frac{x-2}{x+1} \le 0$	$-1 < x \le 2$
intersection	$0 < x \le 2$

A double inequality such as $-3 < \frac{2}{x} < 1$ is a system in disguise: break it into $\frac{2}{x} > -3$ and $\frac{2}{x} < 1$, solve both, intersect. It cannot be solved “all at once” because the middle contains x.

When the answer is empty

$x > 5$ and $x < 2$ have no common value. That is a legitimate answer, written $\varnothing$ or “no solution”, and it must be justified by the picture — not left blank.

THREE SHAPES OF ANSWER

a single interval · a union of intervals · the empty set. Anything else means an error in one of the individual solutions.

Systems are the natural home of the **domain** conditions from Quarter I. “Find the domain of $\frac{\sqrt{x-1}}{x-4}$” is exactly the system $x - 1 \geq 0$ and $x \neq 4$.

02 Worked examples

EXAMPLE 1 **Solve the system $2x - 1 > 3$ and $\frac{1}{x-5} < 0$.**

1. First: $x > 2$.
2. Second: $x < 5$.
   The numerator is positive.
3. Intersect.

ANSWER

$2 < x < 5$

EXAMPLE 2 **Solve $\frac{x}{x-3} \geq 0$ and $x^2 - 16 < 0$.**

1. First: $x \leq 0$ or $x > 3$.
2. Second: $-4 < x < 4$.
3. Intersect both pieces.

ANSWER

$-4 < x \leq 0$ or $3 < x < 4$

EXAMPLE 3 **Find the domain of $f(x) = \frac{\sqrt{x+2}}{x^2-9}$.**

1. Radicand ≥ 0: $x \geq -2$.
2. Denominator $\neq 0$: $x \neq \pm 3$.
3. $x = -3$ is already excluded by the first.

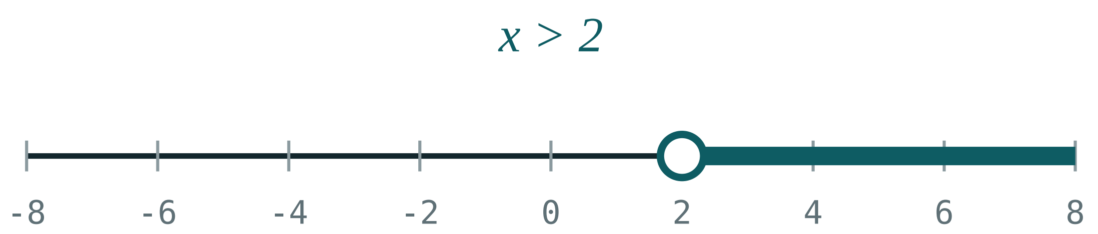
ANSWER

$x \geq -2, x \neq 3$

03 Interactive model

Draw both sets on the board one above the other; the answer is visible before it is written.

Intersecting two solution sets



inequality `x > 2`	boundary `2 (open)`	x = 0 works? **no**	x = 6 works? **yes**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of inequalities	Tengsizliklar sistemasi	Система неравенств
Collection of inequalities	Tengsizliklar majmuasi	Совокупность неравенств
Intersection of sets	To‘plamlar kesishmasi	Пересечение множеств
Union of sets	To‘plamlar birlashmasi	Объединение множеств
Empty solution set	Bo‘sh yechimlar to‘plami	Пустое множество решений
Number line	Sonlar o‘qi	Числовая прямая
Common part	Umumiy qism	Общая часть
Double inequality	Qo‘sh tengsizlik	Двойное неравенство
Bounded interval	Chegaralangan oraliq	Ограниченный промежуток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A system of inequalities asks for:

- the union
- the intersection
- either one
- the difference

$x > 1$ and $x < 4$ gives:

- $1 < x < 4$
- $\mathbb{R}$
- $x > 4$
- $\varnothing$

$x > 5$ and $x < 2$ gives:

- $2 < x < 5$
- all reals
- $\varnothing$
- $x > 5$

$-3 < \frac{2}{x} < 1$ should be:

- solved at once
- split into two inequalities
- cross-multiplied
- squared

The domain of $\frac{\sqrt{x}}{x-1}$ is:

- $x \geq 0$
- $x \geq 0, x \neq 1$
- $x > 1$
- $\mathbb{R}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $x > 2$ and $x < 7$

ANSWER $2 < x < 7$

2. Solve $x \geq 0$ and $x \leq 5$

ANSWER $0 \leq x \leq 5$

3. Solve $x > 4$ and $x < 1$

ANSWER $\varnothing$

4. Solve $2x > 6$ and $x - 1 < 4$

ANSWER $3 < x < 5$

5. Domain of $\dfrac{\sqrt{x}}{x - 2}$

ANSWER $x \geq 0, x \neq 2$

6. Solve $x^2 < 9$

ANSWER $-3 < x < 3$

7. Solve $x > 0$ and $x^2 < 4$

ANSWER $0 < x < 2$

Medium · the standard question

7 PROBLEMS

1. Solve $2x - 1 > 3$ and $\dfrac{1}{x-5} < 0$

ANSWER $2 < x < 5$

2. Solve $\dfrac{x}{x-3} \geq 0$ and $x^2 - 16 < 0$

ANSWER $-4 < x \leq 0$ or $3 < x < 4$

3. Solve $-3 < \dfrac{2}{x} < 1$

ANSWER $x < -\dfrac{2}{3}$ or $x > 2$

4. Domain of $\dfrac{\sqrt{x+2}}{x^2-9}$

ANSWER $x \geq -2, x \neq 3$

5. Solve $\dfrac{1}{x} > 1$ and $x > 0.5$

ANSWER $0.5 < x < 1$

6. Solve $x^2 - 5x + 6 \leq 0$ and $x \neq 2$

ANSWER $2 < x \leq 3$

7. Solve $\dfrac{x-1}{x+2} > 0$ and $x < 3$

ANSWER $x < -2$ or $1 < x < 3$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{1}{x-1} > 2$ and $\frac{1}{x} < 3$

ANSWER $\frac{1}{3} < x < 1.5$ — check ends

2. Solve $x^2 - 4x < 0$ and $\frac{x-3}{x} \geq 0$

ANSWER $3 \leq x < 4$

3. Domain of $\frac{\sqrt{4-x^2}}{x-1}$

ANSWER $-2 \leq x \leq 2, x \neq 1$

4. Domain of $\sqrt{\frac{x-1}{x+2}}$

ANSWER $x < -2$ or $x \geq 1$

5. Find all k for which $x > k$ and $x < 2k$ is non-empty

ANSWER $k > 0$

6. Solve $|x-2| < 3$ and $\frac{1}{x} > 0$

ANSWER $0 < x < 5$

7. Solve $\frac{x^2-1}{x} \geq 0$ and $x \leq 2$

ANSWER $-1 \leq x < 0$ or $1 \leq x \leq 2$

07 Homework

Homework — 6 tasks

Draw one number line per system, with every solution set on it.

1. Solve $3x + 1 > 7$ and $\frac{1}{x-6} < 0$.

2. Solve $\frac{x+1}{x-2} \le 0$ and $x > -3$.
3. Solve $-2 < \frac{3}{x} < 1$.
4. Find the domain of $\frac{\sqrt{x-1}}{x^2-16}$.
5. Find the domain of $\sqrt{\frac{x+3}{x-1}}$.
6. Give a system of two inequalities whose solution set is empty, and prove it with a number line.

Control work 3, and work on the mistakes

Rational equations, systems and inequalities in one paper — then the hour that turns the marks into a plan.

LESSON 38–39 | 2 × 40 MIN | ALGEBRA 10, NAZORAT ISHI 3 | P1 · CHAPTER 1 REVIEW

LESSON CLOCK

3 | 40' | 10' | 25' | 12'

Instructions	3 min
The paper	40 min
Self-mark	10 min
Rewrite	25 min
Common-error drill	12 min

LEARNING OBJECTIVES

- Apply the rational-equation and rational-inequality methods under time.
- State a domain and reject an extraneous root without being prompted.
- Classify each lost mark as careless, method or knowledge.
- Rewrite every wrong solution correctly.

TEXTBOOK

National	pp. 143–146
Cambridge	Pure Mathematics 1, pp. 24–25
Workbook	Control paper C

01 Explanation

The paper — 25 marks, 40 minutes

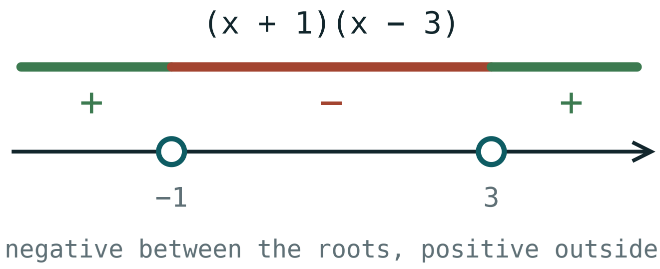
Q	TASK	MARKS	FROM
1	Solve $\frac{3}{x} + \frac{2}{x+1} = 3$, stating the domain	*4*	L28–29
2	Solve $\frac{x^2}{x-5} = \frac{25}{x-5}$ and justify the rejection	*3*	L28–29
3	Solve the system $x + y = 8$, $\frac{1}{x} + \frac{1}{y} = \frac{8}{15}$	*4*	L30–31
4	Solve $\frac{x-4}{x+2} \le 0$ with a sign chart	*4*	L32–34
5	Solve $\frac{2}{x-3} > 1$	*4*	L32–34
6	Solve the system $\frac{x}{x-1} \ge 0$ and $x^2 < 9$	*4*	L35–37
7	Find the domain of $\frac{\sqrt{x+1}}{x^2-4}$	*2*	L35–37

WHERE THE METHOD MARKS ARE

Q1 and Q2 each give one mark for the domain line alone. Q4 and Q5 give two marks for the sign chart, whatever the final interval says. Working, not answers.

The four errors this topic produces

ERROR	LOOKS LIKE	KIND
no domain stated	an extraneous root kept	method
cross-multiplying an inequality	$\frac{1}{x} > 1 \Rightarrow 1 > x$	knowledge
denominator zero included	$x \le 3$ where 3 kills the fraction	method
only one ordered pair	$(3, 5)$ without $(5, 3)$	careless



Questions 4, 5 and 6 in one picture. Two of the marks are for drawing it.

The rewrite

Every question that lost a mark is rewritten in full, with the domain on the first line and the check on the last. Under each rewrite, one sentence beginning *"I lost this because …"*.

> THE DRILL BEFORE THE REWRITE
>
> Ten expressions on the board. For each, the class calls out only **the domain**, in three seconds. Nothing improves this topic faster.

02 Worked examples

EXAMPLE 1 **Model answer, Q2: solve $\frac{x^2}{x-5} = \frac{25}{x-5}$.**

1. Domain $x \neq 5$.
   One mark for this line.
2. $x^2 = 25 \Rightarrow x = \pm 5$
3. $x = 5$ is excluded by the domain.
4. Only $x = -5$ survives.

> ANSWER
>
> $x = -5$

EXAMPLE 2 **Model answer, Q5: solve $\frac{2}{x-3} > 1$.**

1. $\frac{2}{x-3} - 1 > 0$

   Never cross-multiply.

2. $\frac{2-(x-3)}{x-3} = \frac{5-x}{x-3} > 0$

3. Critical 3 and 5, both open.

4. Test $0, 4, 6$: $-, +, -$.

ANSWER

$3 < x < 5$

EXAMPLE 3 **A learner wrote $\frac{x-4}{x+2} \le 0 \Rightarrow -2 \le x \le 4$. Name the mistake.**

1. $x = -2$ makes the denominator zero.

2. A denominator zero is never included.

   Method error.

3. Correct: $-2 < x \le 4$.

ANSWER

Method error — the excluded endpoint was closed

03 Interactive model

Run the domain drill before the rewrite, not after.

Mark these six answers

Domain of $\frac{3}{x} + \frac{2}{x+1}$:

- $x \neq 0$
- $x \neq -1$
- $x \neq 0, -1$
- $\mathbb{R}$

$\frac{x^2}{x-5} = \frac{25}{x-5}$ gives:

- $x = \pm 5$
- $x = 5$
- $x = -5$
- no solution

$\frac{x-4}{x+2} \leq 0$ gives:

- $-2 \leq x \leq 4$
- $-2 < x \leq 4$
- $-2 < x < 4$
- $x \leq 4$

$\frac{2}{x-3} > 1$ gives:

- $x > 3$
- $x < 5$
- $3 < x < 5$
- $x > 5$

The system $x + y = 8$, $\frac{1}{x} + \frac{1}{y} = \frac{8}{15}$ gives:

- $(3,5)$
- $(3,5), (5,3)$
- $(2,6)$
- none

Domain of $\frac{\sqrt{x+1}}{x^2-4}$:

- $x \geq -1$
- $x \geq -1, x \neq 2$
- $x \neq \pm 2$
- $x > -1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Extraneous root	Chet ildiz	Посторонний корень
Sign chart	Ishoralar jadvali	Таблица знаков
Careless error	E’tiborsizlik xatosi	Ошибка по невнимательности
Method error	Usul xatosi	Ошибка в методе
Knowledge gap	Bilim bo‘shlig‘i	Пробел в знаниях
Mark scheme	Baholash sxemasi	Схема оценивания
Correction	Tuzatish	Исправление

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The domain line is worth:

- nothing
- a mark on its own
- the whole question
- half

Cross-multiplying an inequality is:

- fine
- fine if positive
- never allowed with an unknown sign
- always wrong

A denominator zero in a $\leq$ answer is:

- included
- excluded
- sometimes included
- the answer

A symmetric system is finished when you have:

- one pair
- both ordered pairs
- the sum
- the product

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Domain of $\frac{3}{x} + \frac{2}{x+1}$

ANSWER $x \neq 0, -1$

2. Solve $\frac{4}{x} = 2$

ANSWER $x = 2$

3. Solve $\frac{x-4}{x+2} \leq 0$

ANSWER $-2 < x \leq 4$

4. Solve $x^2 < 9$

ANSWER $-3 < x < 3$

5. Solve $x + y = 8, xy = 15$

ANSWER $(3,5), (5,3)$

6. Domain of $\sqrt{x+1}$

ANSWER $x \geq -1$

7. Solve $\frac{1}{x-3} > 0$

ANSWER $x > 3$

Medium · the standard question

7 PROBLEMS

1. Solve $\dfrac{3}{x} + \dfrac{2}{x+1} = 3$

ANSWER $x = \dfrac{3 \pm \sqrt{45}}{6}$ — i.e. $x \approx 1.62, -0.62$

2. Solve $\dfrac{x^2}{x-5} = \dfrac{25}{x-5}$

ANSWER $x = -5$

3. Solve $\dfrac{2}{x-3} > 1$

ANSWER $3 < x < 5$

4. Solve $\dfrac{x}{x-1} \geq 0$ and $x^2 < 9$

ANSWER $-3 < x \leq 0$ or $1 < x < 3$

5. Domain of $\dfrac{\sqrt{x+1}}{x^2-4}$

ANSWER $x \geq -1, x \neq 2$

6. Solve $x + y = 8$, $\dfrac{1}{x} + \dfrac{1}{y} = \dfrac{8}{15}$

ANSWER $(3,5), (5,3)$

7. Solve $\dfrac{x+3}{x-1} < 2$

ANSWER $x < 1$ or $x > 5$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{1}{x-2} + \frac{1}{x+2} = \frac{4}{x^2-4}$

ANSWER no solution — $x = 2$ rejected

2. Solve $\frac{x^2-9}{x^2-4x+3} \geq 0$

ANSWER $x < 1$ or $x \geq 3$, excluding $x = 3$ → $x < 1$ or $x > 3$

3. Solve $x + y = 10$, $x^2 + y^2 = 58$

ANSWER $(3,7), (7,3)$

4. Domain of $\sqrt{\frac{x-2}{x+1}}$

ANSWER $x < -1$ or $x \geq 2$

5. Find k so $\frac{x}{x-k} = 2$ has no solution

ANSWER $k = 0$

6. Solve $\frac{1}{x} > \frac{1}{x-1}$

ANSWER $0 < x < 1$ fails; $x < 0$ or $x > 1$

7. Explain why the answer to Q4 is $-2 < x \leq 4$ and not $-2 \leq x \leq 4$

ANSWER The fraction is undefined at -2

07 Homework

Homework — 4 tasks

Task 1 is the rewrite. It is not optional.

1. Rewrite in full every question that lost a mark, with the domain first and the check last.
2. Five problems from the section your knowledge column was heaviest in.

3. Solve $\frac{5}{x+2} \geq 1$ with a sign chart.

4. Solve $x + y = 12$, $xy = 32$, giving both ordered pairs.

Home / Grade 10 / Algebra / Quarter 2

Irrational equations

Squaring is the only way forward and the only source of false roots — so the check is not optional, it is the method.

LESSON 40–42 | 3 × 40 MIN | ALGEBRA 10, §2.5 | P1 · 1.5 (EXTENSION)

LESSON CLOCK

14' | 24' | 30' | 24' | 36' | 7'

Two conditions, not one	14 min
One radical	24 min
Two radicals	30 min
Substitutions	24 min
Practice	36 min
Homework	7 min

LEARNING OBJECTIVES

- State the two conditions an irrational equation imposes before squaring.
- Solve an equation with one radical by isolating and squaring.
- Solve an equation with two radicals by squaring twice.
- Check every candidate root in the original equation.

TEXTBOOK

National	pp. 147–162
Cambridge	Pure Mathematics 1, pp. 12–17
Workbook	P1 Exercise 1E

01 Explanation

Two conditions before you start

$\sqrt{A} = B$ REQUIRES BOTH

1 $A \geq 0$ — the radicand cannot be negative. **2** $B \geq 0$ — a square root is never negative, so the other side cannot be either. Condition 2 is the one that is forgotten, and it is the one that creates false roots.

$\sqrt{x+3} = -2$ has no solution at all — the left side is never negative — but squaring gives $x + 3 = 4$ and the plausible-looking $x = 1$. Squaring destroys sign information, permanently.

$\sqrt{a} \rightarrow a^{1/2}$

$\sqrt[3]{a} \rightarrow a^{1/3}$

$\sqrt[4]{a} \rightarrow a^{1/4}$

$\sqrt[n]{a^m} \rightarrow a^{m/n}$

a root is a power — the same object, two notations

The graph of a square root lives entirely above the axis. Nothing below it can be an answer.

One radical

1. **Isolate** the radical on one side, alone.
2. Square both sides.
3. Solve the polynomial equation.
4. **Check every root** in the *original* equation.

$$\sqrt{2x + 3} = x \Rightarrow 2x + 3 = x^2 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow x = 3, -1$$

Check: $x = 3$ gives $\sqrt{9} = 3$ ✓. $x = -1$ gives $\sqrt{1} = -1$ ✗. Only $x = 3$.

ISOLATE FIRST, SQUARE SECOND

Squaring $\sqrt{x} + 1 = 4$ directly gives $x + 2\sqrt{x} + 1 = 16$ — a new radical, and more work. Move the 1 across first.

Two radicals

Arrange one radical on each side, square, then isolate whatever radical survives and square again:

$$\sqrt{x + 5} - \sqrt{x} = 1 \Rightarrow \sqrt{x + 5} = 1 + \sqrt{x}$$

$$x + 5 = 1 + 2\sqrt{x} + x \Rightarrow 4 = 2\sqrt{x} \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4$$

Check: $\sqrt{9} - \sqrt{4} = 3 - 2 = 1$ ✓.

WHY ONE ON EACH SIDE

$(a-b)^2$ still contains a cross term with both radicals; $(1+\sqrt{x})^2$ contains only one. Arranging the equation before squaring halves the work.

Substitutions

When the same radical appears more than once, name it:

$$x + \sqrt{x} - 6 = 0\,,\ \text{let } t = \sqrt{x} \ge 0$$

Then $t^2 + t - 6 = 0$, so $t = 2$ or $t = -3$. The second is impossible because $t \ge 0$, so $\sqrt{x} = 2$ and $x = 4$.

The condition $t \ge 0$ is written down with the substitution, not remembered later. It is what removes the false root before any checking is needed.

02 Worked examples

EXAMPLE 1 **Solve $\sqrt{3x+1} = x - 1$.**

1. Conditions: $3x + 1 \ge 0$ and $x - 1 \ge 0$, so $x \ge 1$.
2. $3x + 1 = x^2 - 2x + 1$
3. $x^2 - 5x = 0 \Rightarrow x = 0, 5$
4. $x = 0$ fails $x \ge 1$.

ANSWER

$x = 5$

EXAMPLE 2 **Solve $\sqrt{x+7} - \sqrt{x} = 1$.**

1. $\sqrt{x+7} = 1 + \sqrt{x}$
   One on each side.
2. $x + 7 = 1 + 2\sqrt{x} + x$
3. $6 = 2\sqrt{x} \Rightarrow \sqrt{x} = 3$
4. $x = 9$. Check: $4 - 3 = 1$ ✓

ANSWER

$x = 9$

EXAMPLE 3 **Solve $x - 5\sqrt{x} + 6 = 0$.**

1. Let $t = \sqrt{x} \geq 0$.
   $t^2 - 5t + 6 = 0$
2. $t = 2$ or $t = 3$, both allowed.
3. $\sqrt{x} = 2 \Rightarrow x = 4$
4. $\sqrt{x} = 3 \Rightarrow x = 9$

ANSWER

$x = 4$ and $x = 9$

03 Interactive model

Sketch both sides as graphs — the intersections are the true roots, and false roots are visibly absent.

Roots and powers

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Irrational equation	Irratsional tenglama	Иррациональное уравнение
Radical	Radikal	Радикал
Radicand	Ildiz ostidagi ifoda	Подкоренное выражение
Isolate the radical	Radikalni ajratish	Уединить радикал
Squaring both sides	Kvadratga ko‘tarish	Возведение в квадрат
False (extraneous) root	Chet ildiz	Посторонний корень
Necessary condition	Zaruriy shart	Необходимое условие
Equivalent transformation	Teng kuchli almashtirish	Равносильное преобразование
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\sqrt{A} = B$ requires:

- $A \geq 0$
- $B \geq 0$
- both
- neither

$\sqrt{x + 3} = -2$ has:

- $x = 1$
- $x = -7$
- no solution
- two solutions

The first step with $\sqrt{x} + 1 = 4$ is:

- square
- isolate the radical
- substitute
- cube

For $x + \sqrt{x} - 6 = 0$ set:

- $t = x$
- $t = \sqrt{x}$
- $t = x^2$
- nothing

After squaring you must:

- stop
- check each root in the original
- square again
- factorise

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sqrt{x} = 3$

   ANSWER $x = 9$

2. Solve $\sqrt{x + 1} = 2$

   ANSWER $x = 3$

3. Solve $\sqrt{2x} = 4$

   ANSWER $x = 8$

4. Solve $\sqrt{x} = -1$

   ANSWER no solution

5. Solve $\sqrt{x - 3} = 0$

   ANSWER $x = 3$

6. Solve $\sqrt{5 - x} = 1$

   ANSWER $x = 4$

7. Condition on x in $\sqrt{x - 2}$

   ANSWER $x \geq 2$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{2x + 3} = x$

ANSWER $x = 3$

2. Solve $\sqrt{3x + 1} = x - 1$

ANSWER $x = 5$

3. Solve $\sqrt{x + 7} - \sqrt{x} = 1$

ANSWER $x = 9$

4. Solve $x - 5\sqrt{x} + 6 = 0$

ANSWER $x = 4, 9$

5. Solve $\sqrt{x + 4} = x - 2$

ANSWER $x = 5$

6. Solve $\sqrt{x^2 - 3} = 1$

ANSWER $x = \pm 2$

7. Solve $\sqrt{x} + 1 = 4$

ANSWER $x = 9$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{x + 5} + \sqrt{x} = 5$

ANSWER $x = 4$

2. Solve $\sqrt{2x + 5} - \sqrt{x - 1} = 2$

ANSWER $x = 10$ or $x = 2$

3. Solve $x + \sqrt{x + 5} = 7$

ANSWER $x = 4$

4. Solve $\sqrt{x^2 + 3x} = x + 1$

ANSWER $x = 1$

5. Solve $2x - 7\sqrt{x} + 3 = 0$

ANSWER $x = 9$ or $x = 0.25$

6. Solve $\sqrt{\sqrt{x}} = 2$

ANSWER $x = 16$

7. Find all k for which $\sqrt{x} = x + k$ has exactly one solution

ANSWER $k = 0.25$ or $k \leq 0$

07 Homework

Homework — 6 tasks

Every answer must show the check written out in full.

1. Solve $\sqrt{4x + 1} = x - 1$.
2. Solve $\sqrt{x + 9} - \sqrt{x} = 1$.
3. Solve $x - 7\sqrt{x} + 12 = 0$.
4. Solve $\sqrt{2x - 1} = x - 2$.

5. Explain, in three sentences, why squaring can create a root the original equation never had.
6. Show that $\sqrt{x + 2} = -3$ has no solution without squaring.

Systems of irrational equations

Four lessons on the systems that hide a quadratic — sum and difference of roots, substitution, and the conditions that must survive to the end.

LESSON 43–46 | 4 × 40 MIN | ALGEBRA 10, §2.6 | P1 · 1.5–1.7 (EXTENSION)

LESSON CLOCK

16 | 30' | 30' | 30' | 50' | 24'

Where the difficulty is	16 min
$u = \sqrt{x}$, $v = \sqrt{y}$	30 min
Sum and product of roots	30 min
Mixed systems	30 min
Practice	50 min
Homework and consolidation	24 min

LEARNING OBJECTIVES

- Solve a system of irrational equations by substitution.
- Use $u = \sqrt{x}$, $v = \sqrt{y}$ to turn a system into a polynomial one.
- Track the non-negativity conditions through every step.
- Check every candidate pair in the original system.

TEXTBOOK

National	pp. 163–180
Cambridge	Pure Mathematics 1, pp. 12–23
Workbook	P1 Exercise 1E, 1F

01 Explanation

Where the difficulty is

Nothing new is being asked. A system of irrational equations combines two techniques you already have — the substitution of the last lesson and the elimination of Lesson 30–31. What is new is the **bookkeeping**: two non-negativity conditions instead of one, and they must survive every step.

WRITE THE CONDITIONS ONCE, AT THE TOP

$x \geq 0$, $y \geq 0$, and any others the radicands impose. Then substitute $u = \sqrt{x} \geq 0$ and $v = \sqrt{y} \geq 0$ — and every rejection later is decided by those two inequalities alone.

The standard substitution

$$\sqrt{x} + \sqrt{y} = 5 \text{ and } x + y = 13$$

Put $u = \sqrt{x}$, $v = \sqrt{y}$, both ≥ 0. Then $x = u^2$ and $y = v^2$, and the system becomes polynomial:

$$u + v = 5 \text{ and } u^2 + v^2 = 13$$

So $uv = \frac{25 - 13}{2} = 6$, and u, v are the roots of $t^2 - 5t + 6 = 0$: 2 and 3. Both are non-negative, so both are kept.

> $(x, y) = (4, 9)$ or $(9, 4)$

> SQUARE THE SUBSTITUTION BACK CAREFULLY
>
> $u = 2$ gives $x = 4$, not $x = 2$. Half the errors in this lesson are made in the last line.

Sum and product of the roots

Many systems are stated directly in $\sqrt{x} + \sqrt{y}$ and $\sqrt{xy}$. Since $\sqrt{xy} = \sqrt{x} \cdot \sqrt{y}$ for non-negative x, y, these are exactly $u + v$ and uv:

GIVEN	IN U, V
$\sqrt{x} + \sqrt{y}$	$u + v = s$
$\sqrt{xy}$	$uv = p$
$x + y$	$s^2 - 2p$
$x - y$	$(u - v)(u + v)$

Then u and v are the roots of $t^2 - st + p = 0$, and both must come out non-negative or the pair is rejected.

Mixed systems

When only one equation carries a radical, isolate and substitute in the ordinary way:

> $\sqrt{x + y} = 3$ and $x - y = 1$

The first gives $x + y = 9$ (and requires $x + y \geq 0$, which it now is), so the system is linear: $x = 5$, $y = 4$. Check: $\sqrt{9} = 3$ ✓.

THE FINAL CHECK IS ON THE ORIGINAL SYSTEM

Not on the substituted one. A pair that satisfies $u + v = 5$ may still fail $\sqrt{x} + \sqrt{y} = 5$ if a sign was lost. Substitute the numbers back into the equations as first written.

02 Worked examples

EXAMPLE 1 **Solve $\sqrt{x} + \sqrt{y} = 7, x + y = 25.$**

1. $u + v = 7$, $u^2 + v^2 = 25$, both ≥ 0.
2. $uv = \dfrac{49 - 25}{2} = 12$
3. $t^2 - 7t + 12 = 0 \Rightarrow t = 3, 4$
4. $x = 9, y = 16$ or the reverse.

ANSWER

$(9, 16)$ and $(16, 9)$

EXAMPLE 2 **Solve $\sqrt{x} - \sqrt{y} = 1, x + y = 25.$**

1. $u - v = 1,\ u^2 + v^2 = 25.$
2. $u = v + 1 \Rightarrow (v+1)^2 + v^2 = 25$
3. $2v^2 + 2v - 24 = 0 \Rightarrow v^2 + v - 12 = 0$

   $v = 3$ or $v = -4$ (rejected)
4. $v = 3,\ u = 4$

ANSWER

$(16, 9)$

EXAMPLE 3 **Solve $\sqrt{x + y} = 4, x - y = 2.$**

1. $x + y = 16$

   Condition satisfied.
2. $x - y = 2$
3. Add: $2x = 18 \Rightarrow x = 9.$
4. $y = 7$. Check $\sqrt{16} = 4$ ✓

ANSWER

$(9, 7)$

03 Interactive model

Write the two conditions on the board and keep them visible for the whole solution.

Substituting a root

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64** | index n **3** | power m **1** | check: value^3 **64** | a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
System of irrational equations	Irratsional tenglamalar sistemasi	Система иррациональных уравнений
Non-negativity condition	Manfiy emaslik sharti	Условие неотрицательности
Auxiliary variable	Yordamchi o‘zgaruvchi	Вспомогательная переменная
Sum of roots	Ildizlar yig‘indisi	Сумма корней
Product of roots	Ildizlar ko‘paytmasi	Произведение корней
Back-substitution	Teskari almashtirish	Обратная подстановка
Rejected pair	Rad etilgan juftlik	Отброшенная пара
Symmetric system	Simmetrik sistema	Симметричная система
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In $u = \sqrt{x}$ the condition is:

- $u > 0$
- $u \geq 0$
- $u \leq 0$
- none

If $u = 3$ then x is:

- 3
- 9
- $\sqrt{3}$
- 1.5

$\sqrt{x} + \sqrt{y} = 5$, $x + y = 13$ gives uv:

- 6
- 12
- 13
- 25

A negative value of v means:

- a second answer
- the pair is rejected
- square it
- nothing

The final check is done on:

- the substituted system
- the original system
- neither
- the quadratic

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $\sqrt{x} = 2, y = x + 1$

   ANSWER $(4, 5)$

2. Solve $\sqrt{x + y} = 3, x = y$

   ANSWER $(4.5, 4.5)$

3. If $u = \sqrt{x} = 5$, find x

   ANSWER 25

4. Condition on $u = \sqrt{x}$

   ANSWER $u \geq 0$

5. Solve $\sqrt{x} + \sqrt{y} = 0$

   ANSWER $(0, 0)$

6. Solve $\sqrt{x} = 3, \sqrt{y} = 4$

   ANSWER $(9, 16)$

7. Write $x + y$ in terms of u, v

   ANSWER $u^2 + v^2$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{x} + \sqrt{y} = 5, x + y = 13$

ANSWER $(4,9), (9,4)$

2. Solve $\sqrt{x} + \sqrt{y} = 7, x + y = 25$

ANSWER $(9,16), (16,9)$

3. Solve $\sqrt{x} - \sqrt{y} = 1, x + y = 25$

ANSWER $(16, 9)$

4. Solve $\sqrt{x + y} = 4, x - y = 2$

ANSWER $(9, 7)$

5. Solve $\sqrt{x} + \sqrt{y} = 6, \sqrt{xy} = 8$

ANSWER $(4,16), (16,4)$

6. Solve $\sqrt{x} \cdot \sqrt{y} = 6, x + y = 13$

ANSWER $(4,9), (9,4)$

7. Solve $\sqrt{x - 1} = 2, x + y = 10$

ANSWER $(5, 5)$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{x} + \sqrt{y} = 4, x + y = 10$

   ANSWER $(3 \pm 2\sqrt{2})^2$ — i.e. $u, v = 2 \pm \sqrt{2}$

2. Solve $x + y = 20, \sqrt{x} - \sqrt{y} = 2$

   ANSWER $(16, 4)$

3. Solve $\sqrt{x + y} + \sqrt{x - y} = 4, x^2 - y^2 = 9$

   ANSWER $(5, 4)$

4. Solve $\frac{1}{\sqrt{x}} + \frac{1}{\sqrt{y}} = \frac{5}{6}, xy = 36$

   ANSWER $(4,9), (9,4)$

5. Solve $\sqrt{x} + \sqrt{y} = \sqrt{x + y}$

   ANSWER $x = 0$ or $y = 0$

6. Explain why $\sqrt{x} + \sqrt{y} = 3, x + y = 10$ has no solution

   ANSWER $uv = -0.5 < 0$ is impossible for $u, v \geq 0$

7. Solve $\sqrt{x} + \sqrt{y} = 5, x - y = 5$

   ANSWER $(9, 4)$

07 Homework

Homework — 6 tasks

Write $u \geq 0$ and $v \geq 0$ on the first line of every solution.

1. Solve $\sqrt{x} + \sqrt{y} = 9, x + y = 45$.
2. Solve $\sqrt{x} - \sqrt{y} = 2, x + y = 20$.
3. Solve $\sqrt{x + y} = 5, x - y = 3$.

4. Solve $\sqrt{x} + \sqrt{y} = 8$, $\sqrt{xy} = 15$.
5. Explain why $\sqrt{x} + \sqrt{y} = 2$, $x + y = 10$ has no solution.
6. Check both answers of task 1 by substituting into the original system.

Control work 4, and the quarter review

Irrational equations and systems in one paper, then the map that links a domain to a check to an answer.

LESSON 47–48 | 2 × 40 MIN | ALGEBRA 10, NAZORAT ISHI 4 | P1 · CHAPTER 1 REVIEW

LESSON CLOCK

3 | 42' | 10' | 20' | 10' | 5'

Instructions	3 min
The paper	42 min
Answers	10 min
Rewrite	20 min
Concept map	10 min
Targets	5 min

LEARNING OBJECTIVES

- Apply every method of Quarter II in one assessment.
- Decide which technique a question wants without being told.
- Build a concept map linking domain, transformation and check.
- Set a personal target for Quarter III.

TEXTBOOK

National	pp. 181–184
Cambridge	Pure Mathematics 1, pp. 24–25
Workbook	Control paper D

01 Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Solve $\sqrt{2x+7} = x + 2$, with the conditions stated	5	L40–42
2	Solve $\sqrt{x+5} - \sqrt{x} = 1$	5	L40–42
3	Solve $x - 6\sqrt{x} + 8 = 0$ by substitution	4	L40–42
4	Solve $\sqrt{x} + \sqrt{y} = 6,\ x + y = 20$	6	L43–46
5	Solve $\dfrac{2}{x-1} \geq 1$ with a sign chart	5	L32–34
6	Find the domain of $\dfrac{\sqrt{x-2}}{x^2-25}$	5	L35–37

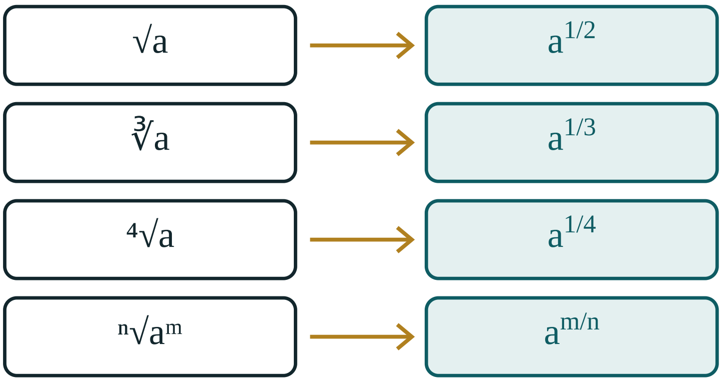
TWO MARKS ARE FOR CONDITIONS

Q1 gives one for $x + 2 \ge 0$ and one for the rejection that follows from it. A correct answer with neither scores three of five.

The concept map

Six boxes, links as sentences:

- **rational equation** → **domain** — “each denominator forbids one value”
- **domain** → **extraneous root** — “multiplying by zero is not reversible”
- **rational inequality** → **sign chart** — “never cross-multiply”
- **system** → **intersection** — “draw them on one line”
- **irrational equation** → **two conditions** — “radicand ≥ 0 and the other side ≥ 0”
- **squaring** → **check** — “sign information is destroyed, so it must be restored”



a root is a power — the same object, two notations

The single picture behind the whole quarter: a root never goes below the axis.

Looking forward

Quarter III opens with the exponential and logarithmic functions. Their equations are solved by exactly the technique of this quarter: state the domain, transform, check. A logarithm demands $argument > 0$ before anything else — the same first line as $\sqrt{A}$ demands $A \ge 0$.

ONE HABIT TO CARRY FORWARD

The first line of every solution states what the unknown is allowed to be. Learners who built that habit this quarter will find Quarter III half-solved before they start.

02 Worked examples

EXAMPLE 1 Model answer, Q1: solve $\sqrt{2x + 7} = x + 2$.

1. Conditions $2x + 7 \geq 0$ and $x + 2 \geq 0$, so $x \geq -2$.
2. $2x + 7 = x^2 + 4x + 4$
3. $x^2 + 2x - 3 = 0 \Rightarrow x = 1, -3$
4. $x = -3$ fails $x \geq -2$.

ANSWER

$x = 1$

EXAMPLE 2 Model answer, Q4: $\sqrt{x} + \sqrt{y} = 6, x + y = 20$.

1. $u + v = 6$, $u^2 + v^2 = 20$, both ≥ 0.
2. $uv = \dfrac{36 - 20}{2} = 8$
3. $t^2 - 6t + 8 = 0 \Rightarrow t = 2, 4$
4. $x = 4, y = 16$ or the reverse.

ANSWER

$(4, 16)$ and $(16, 4)$

EXAMPLE 3 **Model answer, Q6: domain of $\dfrac{\sqrt{x-2}}{x^2-25}$.**

1. $x - 2 \geq 0 \Rightarrow x \geq 2$
2. $x^2 - 25 \neq 0 \Rightarrow x \neq \pm 5$
3. $x = -5$ is already excluded.

ANSWER

$x \geq 2,\ x \neq 5$

03 Interactive model

Work Q1 and Q4 on the board with the conditions written in a box at the top.

The quarter in ten questions

Domain of $\frac{1}{x-4} + \frac{1}{x}$:

- $x \neq 4$
- $x \neq 0$
- $x \neq 0, 4$
- $\mathbb{R}$

$\frac{x^2}{x-3} = \frac{9}{x-3}$ gives:

- ± 3
- 3
- -3
- none

$\frac{1}{x} > 2$ gives:

- $x < 0.5$
- $0 < x < 0.5$
- $x > 0.5$
- $x > 2$

$\frac{x-4}{x+2} \le 0$ gives:

- $-2 \le x \le 4$
- $-2 < x \le 4$
- $-2 < x < 4$
- $x \le 4$

$\sqrt{2x+7} = x + 2$ gives:

- $x = 1$
- $x = 1, -3$
- $x = -3$
- none

$\sqrt{x+5} - \sqrt{x} = 1$ gives:

- $x = 4$
- $x = 5$
- $x = 9$
- $x = 1$

$x - 6\sqrt{x} + 8 = 0$ gives:

$x = 2, 4$

$x = 4, 16$

$x = 16$

$x = 8$

$\sqrt{x} + \sqrt{y} = 6, x + y = 20$ gives:

$(4,16)$ only

$(4,16), (16,4)$

$(2,4)$

none

$\frac{2}{x-1} \geq 1$ gives:

$x \geq 3$

$1 < x \leq 3$

$x \leq 3$

$x > 1$

Domain of $\frac{\sqrt{x-2}}{x^2 - 25}$:

$x \geq 2$

$x \geq 2, x \neq 5$

$x \neq \pm 5$

$x > 2$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Equivalent transformation	Teng kuchli almashtirish	Равносильное преобразование
Non-equivalent step	Teng kuchli bo‘lmagan qadam	Неравносильный переход
Extraneous root	Chet ildiz	Посторонний корень
Sign chart	Ishoralar jadvali	Таблица знаков
Target	Maqsad	Цель
Self-assessment	O‘z-o‘zini baholash	Самооценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first line of any solution this quarter states:

- the answer
- what the unknown may be
- the method
- the check

Squaring is:

- always equivalent
- equivalent only when both sides are ≥ 0
- never allowed
- reversible

Quarter III begins with:

- trigonometry
- the exponential function
- probability
- vectors

A logarithm will demand:

- argument ≥ 0
- argument > 0
- base $= 10$
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sqrt{x} = 6$

ANSWER $x = 36$

2. Domain of $\frac{1}{x - 4}$

ANSWER $x \neq 4$

3. Solve $\frac{3}{x} = 1$

ANSWER $x = 3$

4. Solve $\frac{x - 4}{x + 2} \leq 0$

ANSWER $-2 < x \leq 4$

5. Solve $x + y = 6, xy = 8$

ANSWER $(2,4), (4,2)$

6. Domain of $\sqrt{x - 2}$

ANSWER $x \geq 2$

7. Solve $\sqrt{x + 1} = 3$

ANSWER $x = 8$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{2x + 7} = x + 2$

ANSWER $x = 1$

2. Solve $\sqrt{x + 5} - \sqrt{x} = 1$

ANSWER $x = 4$

3. Solve $x - 6\sqrt{x} + 8 = 0$

ANSWER $x = 4, 16$

4. Solve $\sqrt{x} + \sqrt{y} = 6, x + y = 20$

ANSWER $(4,16), (16,4)$

5. Solve $\frac{2}{x - 1} \geq 1$

ANSWER $1 < x \leq 3$

6. Domain of $\frac{\sqrt{x - 2}}{x^2 - 25}$

ANSWER $x \geq 2, x \neq 5$

7. Solve $\frac{x}{x + 3} = \frac{2}{x}$

ANSWER $x = 6, -1$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{3x + 4} - \sqrt{x + 1} = 1$

   ANSWER $x = 0$ or $x = 8$

2. Solve $\dfrac{x^2 - 1}{x^2 - 4} \leq 0$

   ANSWER $-2 < x \leq -1$ or $1 \leq x < 2$

3. Solve $\sqrt{x} + \sqrt{y} = 5, xy = 36$

   ANSWER $(4,9), (9,4)$

4. Domain of $\sqrt{\dfrac{x - 3}{x + 2}}$

   ANSWER $x < -2$ or $x \geq 3$

5. Solve $2x + \sqrt{x} - 1 = 0$

   ANSWER $x = 0.25$

6. Find k so $\sqrt{x} = x - k$ has exactly two solutions

   ANSWER $0 < k < 0.25$

7. Solve $\dfrac{1}{\sqrt{x}} + \sqrt{x} = 2.5$

   ANSWER $x = 4$ or $x = 0.25$

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. Solve $\sqrt{4x + 5} = x + 2$ and $\sqrt{x + 3} + \sqrt{x} = 3$.

4. Write your target for Quarter III in one checkable sentence, and date it.